

Pixel Art Is a Palette, Not a Downscale: Rebuilding Targets and Decoding for Tiny Sprite Generation

Anonymous CVPR submission

Paper ID ****

Abstract

Text-to-image models fail at the smallest images people draw on purpose: 12–24px RGBA game sprites, where a handful of colours and hard alpha edges are the medium. We show that the usual failure is not a modelling failure but a target failure. In the sprite corpus this area trains on, only 7.1% of the images are natively 16 px or smaller; the rest is 25–64px artwork box-downscaled into the low-resolution buckets, and 4,888 pixel-identical duplicates leave a twin in training for 16.2% of the held-out set. A model fitted to those targets learns soft, fifty-colour blobs, and the evaluation reference, drawn from the same pool, rewards exactly that. We rebuild the targets and evaluate against references that contain only real low-resolution pixel art. Under this protocol the data work alone takes FD-DINOv2 from 167.2 to 117.9 at 16px. We then close the remaining gap where it actually lies: real sprites use a median of six opaque colours while every sampling rule we tested emits more than fifty. Projecting the predicted $\hat{x}_0$ onto a per-sprite k -colour palette during the late denoising steps reaches 31.8 ± 0.5 — better than quantising finished samples (41.6), than one corpus-wide palette (120.1) and than a median filter that removes the same dither (112.2). The rule is a property of the recipe rather than of the network: a transformer backbone that trails the UNet by 36.5 FD without it lands within 0.7 with it. Applied to sprites from generators we did not train, it improves native FD by 36–41% while preserving 80% of the silhouette. Against external systems given the same palette post-process we reach 50.8 versus 134.4 for gpt-image-2 with downscaling.

1. Introduction

Game sprites are among the smallest images anyone draws on purpose: an item or a character on a 12–24px



Figure 1. Four rows of 16px RGBA sprites, nearest-neighbour magnified, with the median number of opaque colours per row. Real sprites drawn at this size use six; the corpus’s own 16px targets, which are 25–64px artwork box-downscaled, use 49; gpt-image-2 rendered large and reduced uses 111; ours uses four. The third row is the easiest on the eye and the furthest from the medium.

RGBA canvas, built from hard edges, a handful of flat colour regions and a transparent background. At these sizes there is no texture in which to hide a mistake — one wrong pixel changes the silhouette, one blended colour reads as blur. The obvious recipe, prompting a large text-to-image model and shrinking its output, produces something recognisable but not a sprite: it keeps the soft shading and the hundred-odd colours of an illustration (Figure 1).

This paper is about a failure one level below the model. We train a dedicated generator for this medium — a 72.5M pixel-space diffusion model over RGBA sprites, conditioned on a frozen CLIP text encoder and a resolution-bucket label — and find its weaknesses inherited from what it is asked to imitate. In the sprite corpus this area trains on, only 7.1% of the images are natively 16px or smaller; two thirds are 25–64px artwork that the pipeline box-downsamples into the low-resolution buckets, and 4,888 pixel-identical duplicates leave a twin in training for 16.2% of the held-out set. A model fitted to those targets learns what they are: soft, fifty-colour blobs. Worse, the standard evaluation

reference is drawn from the same pool, so it rewards the blur it caused, and the ranking of methods under it does not survive contact with real pixel art.

We change both the targets and the yardstick. On the data side we remove the duplicates, forbid feeding a sprite to a bucket more than $1.5\times$ below its native size, and add 14,999 natively small sprites from permissively licensed collections, captioned with a vision-language model. On the evaluation side we define the native protocol: FD-DINOv2 against real sprites that are natively at most R px, never downsampled. At 16 px the data work alone moves the model from 167.2 to 117.9.

What remains is a property of the medium that continuous diffusion cannot express. Real 16 px sprites use a median of six opaque colours and 46% of their adjacent opaque pixel pairs are exactly equal; every sampling rule we tested emits more than fifty colours and reaches 8%, spending the difference on few-level dither inside regions that should be flat. We close that gap inside the sampler: during the late denoising steps we project the predicted $\hat{x}_0$ onto a per-sprite k -colour palette (batched k -means over the opaque pixels, alpha untouched) and continue, so the remaining steps repair the seams hard quantisation leaves. At 16 px this takes FD from 87.1 to 31.8 ± 0.5 , where quantising the finished samples reaches only 41.6.

Two results say the rule is not an artefact of our network. Replacing the UNet with a 69.7M transformer, everything else held fixed, gives a model that trails by 36.5 FD under plain guidance and by 0.7 once the projection is on: the decoding rule absorbs the difference between the two architectures. And the rule applies to sprites we did not generate — re-noising the output of gpt-image-2, FLUX.2 with a pixel-art LoRA, or SDXL with Pixel Art XL and denoising it under our recipe improves native FD by 36–41% at every resolution while keeping about 80% of the input silhouette.

We also report a negative result we consider as informative as the positive one. Cross-resolution self-guidance was, under the downscale-biased reference, our strongest method by 44%; under the native protocol it gives nothing. Which reference set one scores against decides which of these conclusions one publishes.

Contributions.

- A diagnosis of the targets used for tiny-sprite generation — native-resolution statistics, duplicate leakage, and the resulting bias of the standard reference — and a rebuilt corpus plus the native protocol that removes both.
- Palette-projected sampling, which lowers FD-DINOv2 at 16 px from 87.1 to 31.8 ± 0.5 over three

seeds, beats post-hoc quantisation (41.6) and two controls that isolate what it is doing, and costs 0.8 points of caption retrieval at $k=6$, where the samples match the colour statistics of real sprites.

- Evidence that the rule belongs to the recipe rather than the network: it absorbs 98% of the gap between two architectures, and it improves the sprites of three systems we did not train.
- An external comparison in which every system receives the same palette post-process: at 16 px we reach 50.8 against 134.4 for gpt-image-2 with downscaling, 150.1 for FLUX.2 with a pixel-art LoRA and 244.2 for SDXL.

2. The medium and the protocol

What the corpus actually contains. Our starting corpus is the OpenGameArt-derived sprite collection used in this area: 37,489 unique RGBA sprites after cutting sheets into connected components and filtering. Its low-resolution buckets are not made of low-resolution art. Only 7.1% of the sprites are natively 16 px or smaller and 33% are natively 24 px or smaller; two thirds are 25–64 px artwork that the training pipeline box-downsamples into whatever bucket it is asked for. The consequences are measurable in the targets themselves: a real sprite drawn at 16 px has a median of six opaque colours and 46% of its adjacent opaque pixel pairs are exactly equal, while the corpus's own 16 px targets have 49 colours and are soft (Figure 1, rows 1–2).

Duplication compounds it. Hashing every sprite by its exact pixel content finds 4,888 extra copies, and because hold-out sets in this area are defined by path, a pixel-identical twin of a held-out sprite stays in training: 16.2% of the 3,000 evaluation sprites have one, counting a twin anywhere training is allowed to see. We drop every extra copy and every copy of a held-out sprite.

Rebuilt targets. Three changes: remove the duplicates; forbid feeding a sprite to a bucket more than $1.5\times$ below its native size, so a 64 px illustration can no longer become a 16 px target; and add 14,999 natively small sprites cut from three permissively licensed OpenGameArt collections (CC-BY-4.0, CC-BY-SA-3.0, OGA-BY-3.0), captioned with a vision-language model and filtered for fonts, UI elements and near-identical animation frames. Per-file licence records accompany the added material.

The native protocol. The reference set decides the result, so we define it explicitly. Native reference: all

159 real sprites that are natively at most R px, never down-
160 scaled — 2,675 of them at 16 px, split in half so a real-
161 versus-real floor can be computed on a disjoint set (4.47
162 at 16 px). Biased reference: the conventional one, a
163 random 3,000 of the corpus, of which 92.9% is down-
164 scaled artwork at 16 px (floor 3.45). We report both
165 throughout, because the two rank methods differently
166 and the disagreement is part of the result.

167 Scores are Fréchet distances of DINOv2-small CLS
168 features [1, 2] computed at the native resolution:
169 sprites are composited on white and nearest-neighbour
170 upsampled to 224 px. One sample per caption for 3,000
171 held-out captions, which are recaptions of the held-out
172 sprites, so the prompt distribution matches the data.
173 Seed-to-seed standard deviation of our headline con-
174 figuration is 0.49 over three seeds, and an independent
175 retraining of the same recipe lands within 2.5 FD, so we
176 do not read differences below that scale. We also report
177 CLIP ViT-B/32 image-text similarity and retrieval of
178 the own caption among 100, with the distractor draw
179 re-seeded per set.

180 Leakage control. Exact duplicates are removed, but
181 near-duplicates are not: by an 8×8 luminance-plus-
182 alpha average hash, 1,241 of the 3,000 held-out prompts
183 (41.4%) still have a twin in training. Rescoring every
184 configuration on the 1,759 prompts that do not moves
185 each number by at most 3.0 FD — the largest single
186 change is 167.2 to 164.2 — and changes no ordering,
187 so the main tables use the full set.

188 3. Palette-projected sampling

189 4. Experiments

190 5. The rule on other systems

191 6. Discussion and limitations

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